INSTRUCTOR'S DISCUSSION GUIDE

Topic 9: Network Protocols and Services

Follow common network services, compare their functions and security, and inspect transport ports through controlled demonstrations.

Curriculum: Networking 1

Excluded: Title (Slide 1), Objectives (Slide 2), and Summary (Final Slide)

Total Discussion Units: 19 Slides

Slide 3 of 22

Three Core Terms: Protocol, Service, Port

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A protocol defines exchange rules, a service provides a useful function, and a transport port acts like an apartment door number directing traffic to a specific application process.

- Protocol: agreed message formats and behavior, such as Domain Name System (DNS) queries and replies.
- Service: a useful function provided by a running system, such as a Domain Name System (DNS) resolver answering names.
- Port: a Transmission Control Protocol (TCP) or User Datagram Protocol (UDP) endpoint number used to direct traffic to a process; it is not a physical Ethernet socket.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Protocol: agreed message formats and behavior, such as DNS queries and replies.
- Service: a useful function provided by a running system, such as a DNS resolver answering names.
- Port: a TCP or UDP endpoint number used to direct traffic to a process; it is not a physical Ethernet socket.
- A familiar port suggests a convention; it does not prove which program is listening or whether that program is safe.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of three core terms: protocol, service, port.

Key Takeaway: A protocol defines exchange rules, a service provides a useful function, and a transport port acts like an apartment door number directing traffic to a specific application process.

Slide 4 of 22

Four TCP/IP responsibility groups

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

The common Transmission Control Protocol (TCP)/Internet Protocol (IP) model groups network functions into four responsibility areas. Application services depend on transport, Internet delivery, and local-link functions.

- Application: Dynamic Host Configuration Protocol (DHCP), Domain Name System (DNS), Hypertext Transfer Protocol (HTTP), Hypertext Transfer Protocol Secure (HTTPS), and Secure Shell (SSH) provide user-facing and network configuration services; commonly associated with Open Systems Interconnection (OSI) 7.
- Transport: Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) use process ports; commonly associated with Open Systems Interconnection (OSI) 4.
- Internet: Internet Protocol (IP) supplies logical delivery and Internet Control Message Protocol (ICMP) supplies related diagnostics; commonly associated with Open Systems Interconnection (OSI) 3.
- Link: local framing and physical transmission; Address Resolution Protocol (ARP) supports Internet Protocol version 4 (IPv4) neighbor resolution at this local-link boundary.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Application: DHCP, DNS, HTTP, HTTPS, and SSH provide user-facing and network configuration services; commonly associated with OSI 7.
- Transport: TCP and UDP use process ports; commonly associated with OSI 4.
- Internet: IP supplies logical delivery and ICMP supplies related diagnostics; commonly associated with OSI 3.
- Link: local framing and physical transmission; ARP supports IPv4 neighbor resolution at this local-link boundary.
- OSI/TCP-IP mappings are approximate.
- Advanced enterprise protocols like SNMP, NTP, and dedicated mail servers are studied in Networking II.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of four tcp/ip responsibility groups.

Key Takeaway: The common Transmission Control Protocol (TCP)/Internet Protocol (IP) model groups network functions into four responsibility areas. Application services depend on transport, Internet delivery, and local-link functions.

Slide 5 of 22

Two transport services: TCP and UDP

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Transmission Control Protocol supplies ordered reliable bytes, while User Datagram Protocol carries independent datagrams without built-in delivery guarantees. Their differences affect how application protocols handle communication.

- Transmission Control Protocol (TCP): connection-oriented, ordered reliable bytes, with acknowledgments and retransmission. Useful when the application needs this stream service.
- User Datagram Protocol (UDP): connectionless datagrams without built-in delivery or ordering guarantees. Useful when a small exchange or application-controlled behavior fits the task.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- TCP: connection-oriented, ordered reliable bytes, with acknowledgments and retransmission.
- Useful when the application needs this stream service.
- UDP: connectionless datagrams without built-in delivery or ordering guarantees.
- Useful when a small exchange or application-controlled behavior fits the task.
- Neither “UDP is always faster” nor “TCP proves the application processed the request” is a reliable conclusion.
- Application protocols can add recovery above UDP.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of two transport services: tcp and udp.

Key Takeaway: Transmission Control Protocol supplies ordered reliable bytes, while User Datagram Protocol carries independent datagrams without built-in delivery guarantees. Their differences affect how application protocols handle communication.

Slide 6 of 22

Client and Server Endpoints

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

A client commonly uses a temporary source port to contact a server's listening port. Addresses, ports, and transport together identify the communication context.

- A normal client often uses a temporary source port, such as Transmission Control Protocol (TCP) 53001, to contact a server listening on TCP 443. The response returns to that client endpoint.
- The connection context includes transport plus source/destination addresses and ports. Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) can use the same numeric port independently.
- System ports are 0–1023, user/registered ports 1024–49151, and dynamic/private ports 49152–65535. Port 0 is reserved; operating systems may choose temporary ports from implementation-specific ranges.
- A reachable Internet Protocol (IP), an open transport endpoint, Transport Layer Security (TLS) validation, and a working application are separate observations. Testing one does not prove all the others.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌞 REAL-WORLD ANALOGY / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- A normal client often uses a temporary source port, such as TCP 53001, to contact a server listening on TCP 443.
- The response returns to that client endpoint.
- The connection context includes transport plus source/destination addresses and ports.
- TCP and UDP can use the same numeric port independently.
- System ports are 0–1023, user/registered ports 1024–49151, and dynamic/private ports 49152–65535.
- Port 0 is reserved; operating systems may choose temporary ports from implementation-specific ranges.
- A reachable IP, an open transport endpoint, TLS validation, and a working application are separate observations.
- Testing one does not prove all the others.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of client and server endpoints.

Key Takeaway: A client commonly uses a temporary source port to contact a server's listening port. Addresses, ports, and transport together identify the communication context.

Slide 7 of 22

Transport Reliability: TCP vs UDP

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

Transport protocols differ in their response to missing data. Transmission Control Protocol provides recovery and ordered bytes, while User Datagram Protocol leaves any needed recovery to the application.

Interactive Demonstration: `osiLayer` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Compare TCP and UDP using the same modeled missing data unit.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of transport reliability: tcp vs udp.

Key Takeaway: Transport protocols differ in their response to missing data. Transmission Control Protocol provides recovery and ordered bytes, while User Datagram Protocol leaves any needed recovery to the application.

Slide 8 of 22

Internet Protocol (IP)

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Internet Protocol supplies logical addressing and packet delivery across networks. It provides best-effort delivery rather than an ordered reliable application stream.

- Practical application: Internet Protocol version 4 (IPv4) or Internet Protocol version 6 (IPv6) identifies source/destination interfaces and supports router forwarding.
- Internet Protocol (IP) provides best-effort delivery, not a reliable application byte stream. It has no Transmission Control Protocol (TCP)/User Datagram Protocol (UDP) port. Internet Protocol version 6 (IPv6) uses 128-bit addresses and does not use Address Resolution Protocol (ARP).

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Practical application: IPv4 or IPv6 identifies source/destination interfaces and supports router forwarding.
- IP provides best-effort delivery, not a reliable application byte stream.
- It has no TCP/UDP port.
- IPv6 uses 128-bit addresses and does not use ARP.
- OSI 3 · TCP/IP Internet.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of internet protocol (ip).

Key Takeaway: Internet Protocol supplies logical addressing and packet delivery across networks. It provides best-effort delivery rather than an ordered reliable application stream.

Slide 9 of 22

Address Resolution Protocol (ARP)

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Address Resolution Protocol resolves the hardware address of an Internet Protocol version 4 (IPv4) next hop on a local link. A remote destination usually requires the gateway's local hardware address rather than the remote host's.

- Practical application: Find a local peer's Ethernet Media Access Control (MAC), or the gateway MAC when sending toward a remote Internet Protocol (IP) network.
- Address Resolution Protocol (ARP) is not carried inside Transmission Control Protocol (TCP) or User Datagram Protocol (UDP) and has no port. Ethernet ARP uses EtherType 0x0806. Routers do not forward a local ARP broadcast across Internet Protocol (IP) networks; Internet Protocol version 6 (IPv6) uses Neighbor Discovery instead.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌞 REAL-WORLD ANALOGY / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Practical application: Find a local peer's Ethernet MAC, or the gateway MAC when sending toward a remote IP network.
- ARP is not carried inside TCP or UDP and has no port.
- Ethernet ARP uses EtherType 0x0806.
- Routers do not forward a local ARP broadcast across IP networks; IPv6 uses Neighbor Discovery instead.
- Local-link boundary · TCP/IP Link; commonly associated with OSI 2.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of address resolution protocol (arp).

Key Takeaway: Address Resolution Protocol resolves the hardware address of an Internet Protocol version 4 (IPv4) next hop on a local link. A remote destination usually requires the gateway's local hardware address rather than the remote host's.

Slide 10 of 22

How ARP Resolves MAC Addresses

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

A host needs a local delivery address before sending an Ethernet frame toward an Internet Protocol version 4 (IPv4) neighbor. Address Resolution Protocol uses a local request and reply to establish that mapping.

Interactive Demonstration: `protocolArp` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Change between a local peer and a remote destination; follow the calculated next-hop explanation.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of how arp resolves mac addresses.

Key Takeaway: A host needs a local delivery address before sending an Ethernet frame toward an Internet Protocol version 4 (IPv4) neighbor. Address Resolution Protocol uses a local request and reply to establish that mapping.

Slide 11 of 22

ICMP and Network Ping Basics

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Internet Control Message Protocol carries diagnostic and error information related to Internet Protocol delivery. Echo messages and expiry reports provide evidence without proving that every application service works.

- Practical application: Ping commonly uses Echo; traceroute-style tools use Time to Live (TTL)/hop-limit behavior and returned diagnostics.
- Internet Control Message Protocol (ICMP) has no Transmission Control Protocol (TCP)/User Datagram Protocol (UDP) port. ICMP for Internet Protocol version 4 (IPv4) uses Internet Protocol (IP) protocol number 1; Internet Control Message Protocol for IPv6 (ICMPv6) uses next-header 58. Missing replies may mean filtering or loss. It does not fix a failed link.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

☀️ REAL-WORLD ANALOGY / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Practical application: Ping commonly uses Echo; traceroute-style tools use TTL/hop-limit behavior and returned diagnostics.
- ICMP has no TCP/UDP port.
- ICMP for IPv4 uses IP protocol number 1; ICMPv6 uses next-header 58.
- Missing replies may mean filtering or loss.
- It does not fix a failed link.
- OSI 3 · TCP/IP Internet.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of icmp and network ping basics.

Key Takeaway: Internet Control Message Protocol carries diagnostic and error information related to Internet Protocol delivery. Echo messages and expiry reports provide evidence without proving that every application service works.

Slide 12 of 22

Dynamic Host Configuration Protocol (DHCP)

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Dynamic Host Configuration Protocol supplies leased host configuration and related options. Clients use it to obtain settings such as address, mask, gateway, and name resolver.

- Practical application: Automatically configure classroom PCs without entering each address manually.
- Initial allocation is explained as Discover, Offer, Request, Acknowledgment (ACK). The ports below describe DHCPv4; DHCPv6 has different messages and ports.
- User Datagram Protocol (UDP) 67: DHCPv4 server.
- User Datagram Protocol (UDP) 68: DHCPv4 client.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Practical application: Automatically configure classroom PCs without entering each address manually.
- Initial allocation is explained as Discover, Offer, Request, ACK.
- The ports below describe DHCPv4; DHCPv6 has different messages and ports.
- UDP 67: DHCPv4 server.
- UDP 68: DHCPv4 client.
- OSI 7 · TCP/IP Application.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of dynamic host configuration protocol (dhcp).

Key Takeaway: Dynamic Host Configuration Protocol supplies leased host configuration and related options. Clients use it to obtain settings such as address, mask, gateway, and name resolver.

Slide 13 of 22

DHCP Ports and Communication

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

Dynamic Host Configuration Protocol version 4 uses server User Datagram Protocol port 67 and client port 68. Its initial exchange proposes, requests, and confirms an address configuration.

Interactive Demonstration: dhcp — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- DHCPv4 uses server UDP 67 and client UDP 68; a lease becomes active at ACK.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of dhcp ports and communication.

Key Takeaway: Dynamic Host Configuration Protocol version 4 uses server User Datagram Protocol port 67 and client port 68. Its initial exchange proposes, requests, and confirms an address configuration.

Slide 14 of 22

Domain Name System (DNS)

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

The Domain Name System answers queries about names using records such as host addresses and mail destinations. Name resolution supplies information for later communication rather than carrying the application traffic itself.

- Practical application: Find the address for a school web or mail server by name.
- Classic Domain Name System (DNS) supports User Datagram Protocol (UDP) and Transmission Control Protocol (TCP) on port 53. TCP is not limited to zone transfers: it also supports other queries, including fallback after truncation. Caches respect record lifetimes. Encrypted DNS has additional transports.
- User Datagram Protocol (UDP) 53: Classic Domain Name System (DNS) queries/responses.
- Transmission Control Protocol (TCP) 53: Classic Domain Name System (DNS) over a TCP connection.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Practical application: Find the address for a school web or mail server by name.
- Classic DNS supports UDP and TCP on port 53.
- TCP is not limited to zone transfers: it also supports other queries, including fallback after truncation.
- Caches respect record lifetimes.
- Encrypted DNS has additional transports.
- UDP 53: Classic DNS queries/responses.
- TCP 53: Classic DNS over a TCP connection.
- OSI 7 · TCP/IP Application.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of domain name system (dns).

Key Takeaway: The Domain Name System answers queries about names using records such as host addresses and mail destinations. Name resolution supplies information for later communication rather than carrying the application traffic itself.

Slide 15 of 22

DNS Caching and Record Lifetimes

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

A cached name answer can be reused until its permitted lifetime expires. Server record changes may therefore remain unseen by a client until its cache entry is refreshed.

Interactive Demonstration: `dnsCacheDemo` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *Follow common network services, compare their functions and security, and inspect transport ports through controlled demonstrations.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Resolve names, change a server record, and advance the client cache lifetime.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of dns caching and record lifetimes.

Key Takeaway: A cached name answer can be reused until its permitted lifetime expires. Server record changes may therefore remain unseen by a client until its cache entry is refreshed.

Slide 16 of 22

Hypertext Transfer Protocol (HTTP)

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Hypertext Transfer Protocol defines requests and responses for web resources. Methods, status codes, and message content describe the requested operation and its result.

- Practical application: A browser requests a page or a client calls a web Application Programming Interface (API).
- Ordinary Hypertext Transfer Protocol (HTTP) does not provide Transport Layer Security (TLS) encryption. HTTP/1.1 commonly uses Transmission Control Protocol (TCP); port 80 is the conventional HTTP service endpoint. An HTTP status error differs from a failure to reach the server.
- Transmission Control Protocol (TCP) 80: Conventional unencrypted Hypertext Transfer Protocol (HTTP) service.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Practical application: A browser requests a page or a client calls a web API.
- Ordinary HTTP does not provide TLS encryption.
- HTTP/1.1 commonly uses TCP; port 80 is the conventional HTTP service endpoint.
- An HTTP status error differs from a failure to reach the server.
- TCP 80: Conventional unencrypted HTTP service.
- OSI 7 · TCP/IP Application.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of hypertext transfer protocol (http).

Key Takeaway: Hypertext Transfer Protocol defines requests and responses for web resources. Methods, status codes, and message content describe the requested operation and its result.

Slide 17 of 22

HTTPS and Secure Web Browsing

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Hypertext Transfer Protocol Secure protects web communication using Transport Layer Security. Certificate validation and authenticated encryption protect the connection without guaranteeing that the website's content is trustworthy.

- Practical application: Access a protected classroom portal without exposing application content on the local path.
- Hypertext Transfer Protocol (HTTP)/1.1 and HTTP/2 commonly use Transport Layer Security (TLS) over Transmission Control Protocol (TCP) 443. HTTP/3 uses QUIC with TLS security over User Datagram Protocol (UDP), commonly 443. Hypertext Transfer Protocol Secure (HTTPS) does not prove a site is trustworthy or hide every piece of network metadata.
- Transmission Control Protocol (TCP) 443: Hypertext Transfer Protocol Secure (HTTPS) using Hypertext Transfer Protocol (HTTP)/1.1 or HTTP/2.
- User Datagram Protocol (UDP) 443: Hypertext Transfer Protocol (HTTP)/3 over QUIC; common Hypertext Transfer Protocol Secure (HTTPS) endpoint.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Practical application: Access a protected classroom portal without exposing application content on the local path.
- HTTP/1.1 and HTTP/2 commonly use TLS over TCP 443.
- HTTP/3 uses QUIC with TLS security over UDP, commonly 443.
- HTTPS does not prove a site is trustworthy or hide every piece of network metadata.
- TCP 443: HTTPS using HTTP/1.1 or HTTP/2.
- UDP 443: HTTP/3 over QUIC; common HTTPS endpoint.
- OSI 7 service · TCP/IP Application; TLS spans security/representation responsibilities.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of https and secure web browsing.

Key Takeaway: Hypertext Transfer Protocol Secure protects web communication using Transport Layer Security. Certificate validation and authenticated encryption protect the connection without guaranteeing that the website's content is trustworthy.

Slide 18 of 22

Example: open a protected school website

EXAMPLE

📖 WHAT TO READ & DISPLAY ON SCREEN

Opening a protected website combines name resolution with a secure application exchange. The resolver supplies an address, while the browser exchanges web traffic with the server.

- Domain Name System (DNS) resolves the server name. For a common Hypertext Transfer Protocol (HTTP)/1.1-over-TLS example, Transmission Control Protocol (TCP) connects to server port 443 and Transport Layer Security (TLS) protects the exchange.
- Hypertext Transfer Protocol Secure (HTTPS) carries the web request and response inside that protected connection. The resolver is not a relay for the web page.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *Follow common network services, compare their functions and security, and inspect transport ports through controlled demonstrations.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Domain Name System (DNS) resolves the server name.
- For a common HTTP/1.1-over-TLS example, Transmission Control Protocol (TCP) connects to server port 443 and Transport Layer Security (TLS) protects the exchange.
- Hypertext Transfer Protocol Secure (HTTPS) carries the web request and response inside that protected connection.
- The resolver is not a relay for the web page.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of example: open a protected school website.

Key Takeaway: Opening a protected website combines name resolution with a secure application exchange. The resolver supplies an address, while the browser exchanges web traffic with the server.

Slide 19 of 22

How Protocols Cooperate on the Web

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

A web request depends on local delivery, name resolution, transport, connection security, and the web service. A failure at one stage produces different evidence from a failure at another.

Interactive Demonstration: `protocolWeb` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Follow ARP, classic DNS, IP, TCP, TLS, and HTTP; change a condition to see exactly where the operation differs.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of how protocols cooperate on the web.

Key Takeaway: A web request depends on local delivery, name resolution, transport, connection security, and the web service. A failure at one stage produces different evidence from a failure at another.

Slide 20 of 22

Secure Shell (SSH)

CONCEPT

📖 WHAT TO READ & DISPLAY ON SCREEN

Secure Shell provides authenticated encrypted remote access and channels. Checking server identity and user authorization is part of establishing a trusted connection.

- Practical application: Administer an authorized router/server or carry an encrypted administrative session.
- Check the server host identity and authenticate using an authorized account. A port number alone does not establish that authentication or encryption is correctly configured.
- Transmission Control Protocol (TCP) 22: Conventional Secure Shell (SSH) server endpoint.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Practical application: Administer an authorized router/server or carry an encrypted administrative session.
- Check the server host identity and authenticate using an authorized account.
- A port number alone does not establish that authentication or encryption is correctly configured.
- TCP 22: Conventional SSH server endpoint.
- OSI 7 · TCP/IP Application.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of secure shell (ssh).

Key Takeaway: Secure Shell provides authenticated encrypted remote access and channels. Checking server identity and user authorization is part of establishing a trusted connection.

Slide 21 of 22

Common Protocols and Port Numbers

SIMULATION

📖 WHAT TO READ & DISPLAY ON SCREEN

Protocol and port references identify conventional service roles and endpoint bindings. The transport must accompany the number, and some network protocols have no transport port at all.

Interactive Demonstration: `protocolDirectory` — Use on-screen controls to step through or demonstrate the concept visually.

💡 BRIEF EXPLANATION AFTER READING (TEACHER'S DISCUSSION NOTES)

🌟 REAL-WORLD ANALOGY / INTUITION

> *A classroom laptop obtains configuration, looks up the school server, downloads a protected handout, and connects securely via SSH.*

🗣️ HOW TO EXPLAIN IT (TALKING POINTS)

- Inspect all ten core protocols individually, including transport and layer relationships.

❓ QUICK CHECK FOR UNDERSTANDING (ASK THE CLASS)

Teacher Asks: "Ask the class: Why is this concept essential when setting up or troubleshooting a modern network?"

Expected Answer / Guidance: Encourages learners to explain the functional role and real-world impact of common protocols and port numbers.

Key Takeaway: Protocol and port references identify conventional service roles and endpoint bindings. The transport must accompany the number, and some network protocols have no transport port at all.